Università di Pisa

Mott localisation, antiferromagnetism and nematic superconductivity in the extended Hubbard model with non-local pairing

Candidato:
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Outline

Cuprates and Hubbard physics

Superconductivity in cuprates

Nobel Prize in Physics 1987

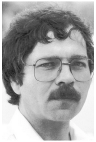
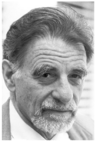


Photo from the Nobel Foundation archive.

J. Georg Bednorz

Prize share: 1/2



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K. Alexander Müller

Prize share: 1/2

Possible High T_c Superconductivity in the Ba – La – Cu – O System

J.G. Bednorz and K.A. Müller
IBM Zürich Research Laboratory, Rüschlikon, Switzerland

Received April 17, 1986

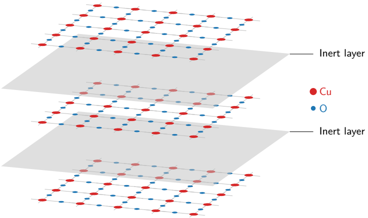
What makes cuprates interesting?

- Strong "unconventional" insulators;
- Rich phase diagram;
- Origin of Cooper pairing: still unclear;
- Highest critical temperatures for SC transition in this class of materials.

Cuprates and the Hubbard model

These materials are stackings of:

- 2D copper oxide active layers;
- intermediate layers of inert material.



Most of the cuprates physics is well described by the single-band Hubbard model on the square lattice:

$$\hat{H}_{\mathrm{HM}} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma}}_{\text{Kinetic operator}} \underbrace{+U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\text{On-site repulsion}} \qquad \text{with } t, U > 0.$$

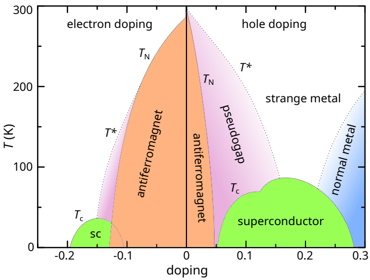
Within cuprates phase diagram:

- **Antiferromagnetism** at null and low doping;
- d-**wave superconductivity** as we inject or remove charge.

Both arise from **strong electronic correlations**.

$\Downarrow$

Failure of band theory in describing insulation.



Schematic phase diagram of cuprates. Author: Holger Motzkau, Wikimedia commons, CC BY-SA 3.0.

What is the origin of the insulation mechanism?

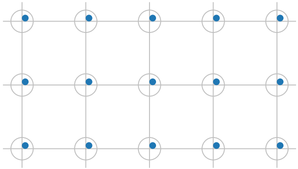
Strong e-e interactions and **big energetic cost** of having two electrons on the same site.

Strongly coupled Hubbard model:

$$U \gg t\,.$$

$$\Downarrow$$

Mott insulation mechanism
Electrons are **localised** as in a sort of "traffic jam".



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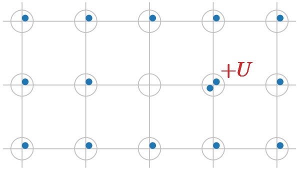


Mott physics, antiferromagnetism and d-wave superconductivity

How does Mott localisation relate to antiferromagnetism and superconductivity?

Antiferromagnetism

Mott localisation at half-filling.

$\Downarrow$

Singlet pairing $\leftrightarrow$ AF ordering.

$\Downarrow$

HM at strong coupling:

$$\hat{H}_{\mathrm{HM}} \simeq \frac{4t^2}{U} \sum_{\langle ij \rangle} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j + \Delta E \,,$$

Superconductivity

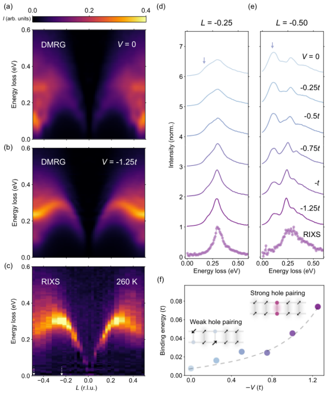
Condensation of Cooper pairs with d-wave spatial structure.

$\Downarrow$

Order parameter $\psi(\mathbf{r}_1 - \mathbf{r}_2)$: antisymmetric upon $\pi/2$ rotations.

$\Downarrow$

Suppressed on-site paring.



Hari Padma et al., 2025. **Physical Review X 15 (2): 021049**.

Incompatibilities with the spectrum of magnetic excitation in cuprates.

$$\Downarrow$$

Proposal: extension of the model with a **non-local attraction**,

$$\hat{H}_{\mathrm{EHM}} = \hat{H}_{\mathrm{HM}} \underbrace{-V \sum_{\langle ij \rangle} \hat{n}_i \hat{n}_j}_{\hat{H}_V} \, .$$

$\hat{H}_V$ can act **as a source of d-wave superconductivity**.

The model (EHM):

$$\hat{H}_{\mathrm{EHM}} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma}}_{\text{Kinetic operator}} \underbrace{+U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\text{Local repulsion}} \underbrace{-V \sum_{\langle ij \rangle} \hat{n}_i \hat{n}_j}_{\text{Non-local attraction}} \qquad \text{with } t, U, V > 0.$$

The motivation: Compatibility with recent experimental findings.

The idea: we know that the EHM can host d-wave superconductivity. Can we obtain Mott physics?

The method: theoretical and numerical Hartree-Fock technique.

The Hartree-Fock method

The constrained minimisation framework

We want to approximate the true ground state $|\Psi_0\rangle$ of the EHM, with energy E_0.

Variational principle:

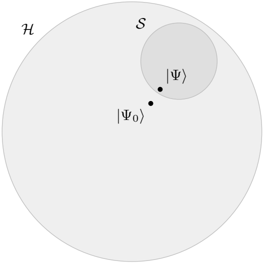
$$\forall\, |\Phi\rangle \in \mathcal{S} \subset \mathcal{H} \quad : \quad E[\Phi] \geq E_0\,.$$

$$\Downarrow$$

Constrained minimisation procedure

$$|\Psi\rangle \quad : \quad \left.\frac{\delta E[\Phi]}{\delta\,|\Phi\rangle}\right|_{|\Phi\rangle = |\Psi\rangle} = 0\,.$$

The state $|\Psi\rangle$ is the best approximation to $|\Psi_0\rangle$ inside of $\mathcal{S}$.



Sketch of the Hilbert space $\mathcal{H}$ and a subset $\mathcal{S}$.

Hartree-Fock method: $\mathcal{S}$ is the set of all **gaussian states**, i.e. states that are non-interacting for some fermionic operators

$$|\Psi\rangle = \bigotimes_{\alpha} \hat{\gamma}_\alpha^\dagger \, |\Omega_\gamma\rangle \ , \qquad \left\{\hat{\gamma}_\alpha, \hat{\gamma}_\beta^\dagger\right\} = \delta_{\alpha\beta} \ ,$$

with $\hat{\gamma} \leftrightarrow \hat{c}$ through a unitary map.

$$\Downarrow$$

Wick's theorem: averages of quartic operators decompose nicely,

$$\langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_\gamma \hat{c}_\delta \rangle = \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\delta \rangle \langle \hat{c}_\beta^\dagger \hat{c}_\gamma \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\gamma \rangle \langle \hat{c}_\beta^\dagger \hat{c}_\delta \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \rangle \langle \hat{c}_\gamma \hat{c}_\delta \rangle}_{\text{Cooper}} \, .$$

The EHM hamiltonian is a quartic operator!

The self-consistency mapping

Each $|\Psi\rangle \in \mathcal{S}$ is uniquely defined by a particular choice of the **variational parameters** (HFPs)

$$\Delta^{+-}_{\alpha\beta} = \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle \,, \qquad \Delta^{++}_{\alpha\beta} = \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \,.$$

Let $\mathbf{v}$ be the vector that collects all HFPs. We aim to find

$$|\Psi[\mathbf{v}]\rangle \simeq |\Psi_0\rangle \ ,$$

with $\mathbf{v}$ such that:

Variational picture

Energy is minimised,

$$\left.\frac{\delta E[\Phi]}{\delta\, |\Phi\rangle}\right|_{|\Phi\rangle = |\Psi[\mathbf{v}]\rangle} = 0 \,.$$

Self-consistency picture

A self-consistency problem is satisfied,

$$\mathrm{A}(\mathbf{v}) = \mathbf{v} \,,$$

being A a non-linear map **uniquely determined**.

The HF method and symmetry breaking mechanisms

Phase transition: spontaneous breaking of some symmetries of the model.

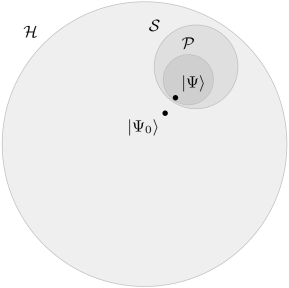
Phase X: shows symmetries Y, Z ...

$$\Downarrow$$

Further constraining of HF subspace to $\mathcal{P} \subset \mathcal{S}$, states with symmetries Y, Z ...

$$\Downarrow$$

Symmetry selection rules on the variational parameters $\mathbf{v}$.



e.g.

$$\text{Charge is conserved} \quad \Longrightarrow \quad \langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \rangle = 0$$

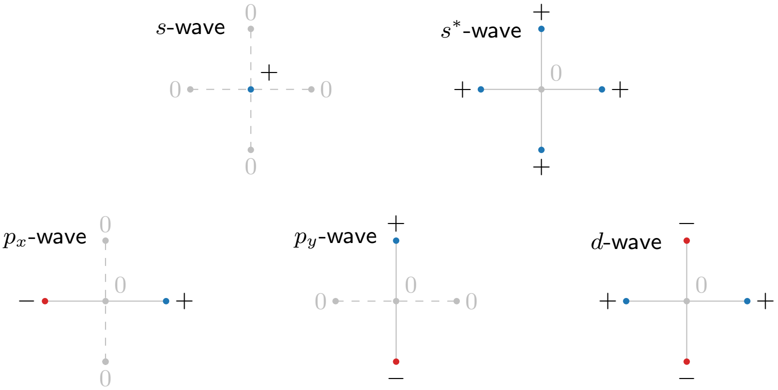
The model (EHM):

$$\hat{H}_{\mathrm{EHM}} = \underbrace{-t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}}_{\text{Kinetic operator}} \underbrace{+U\sum_{i}\hat{n}_{i\uparrow}\hat{n}_{i\downarrow}}_{\text{Local repulsion}} \underbrace{-V\sum_{\langle ij\rangle}\hat{n}_{i}\hat{n}_{j}}_{\text{Non-local attraction}} \qquad \text{with } t, U, V > 0,$$

and its symmetries:

- *Gauge invariance* (charge conservation), $\mathrm{U}(1)$.
  If broken: superconductivity.
- *Global spin rotational invariance*, $\mathrm{SU}(2)$.
  If broken: magnetism.
- *Translational invariance* in directions x and y, $\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$.
  If broken: non-uniform charge and spin densities.
- *Rotational invariance* under $\pi/2$ rotations, C_4.
  If broken: nematicity.

It is useful to introduce five symmetry structures:



The normal phase

Normal phase: no broken symmetries. From the Fock channel,

$$\tilde{t}_{ij\sigma} \equiv \begin{cases} t - V\langle \hat{c}^{\dagger}_{j\sigma}\hat{c}_{i\sigma}\rangle & \text{if } i, j \text{ are NNs,} \\ 0 & \text{otherwise.} \end{cases}$$

$$\Downarrow$$

Self-consistency problem: one HFP,

$$u^{(s^*)} = \left(s^*\text{-wave component of } u_{\mathbf{k}} \equiv \langle \hat{n}_{\mathbf{k}}\rangle\right),$$

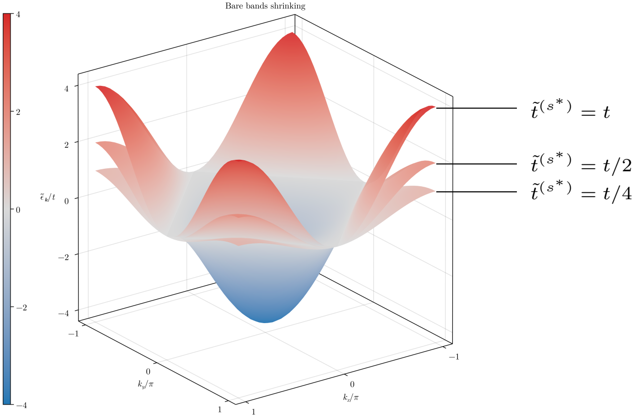
s.t. hopping amplitude is isotropically renormalised,

$$t \quad \rightarrow \quad \tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)}V}{2}\,.$$

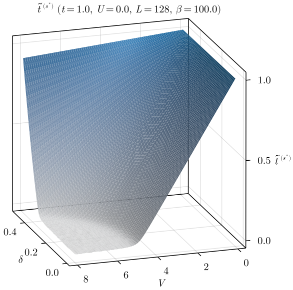
Therefore: renormalisation of electronic bands, $\epsilon_{\mathbf{k}\sigma} \rightarrow \tilde{\epsilon}_{\mathbf{k}\sigma}$.

Theoretical constraint:

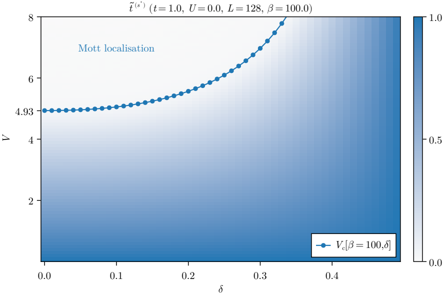
$0 < \tilde{t}^{(s^*)} < t \quad \Longrightarrow$ **Flattening** of bands = Mott localisation.



In (δ, V) plane, we have determined theoretically the presence of a Mott critical boundary $V_{\mathrm{c}}[\beta, \delta]$,



(a) Numerical results for $\tilde{t}^{(s^*)}$



(b) Critical Mott boundary.

Within the Mott plateau, electrons are completely localised.

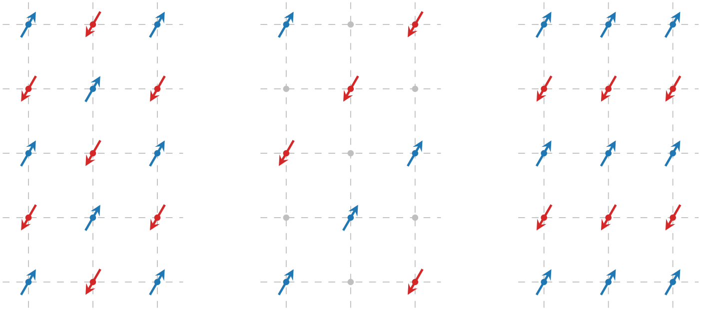
The HF solution in the normal phase

The **normal phase**, summarised:

- Preserves all symmetries of the EHM;
- The HF approximation is **independent of** U;
- Presence of V-induced Mott localisation and **bands flattening**;
- We identify a Mott transition boundary as we vary the doping level.

The antiferromagnetic phase

Among all possible realisations of AF ordering,



Alternating

Doubled Periodicity

Stripes

...

we restrict to "commensurate antiferromagnetism".

Commensurate antiferromagnetism

Among all possible realisations of AF ordering,

we restrict to “commensurate antiferromagnetism”.

Antiferromagnetism in the conventional Hubbard model

AF phase: lowers $\mathrm{SU}(2)$ and $\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$.

In the HM ($V = 0$), the only HFP is the magnetisation m.

$$\Downarrow$$

U-induced magnetisation, disrupted by δ.

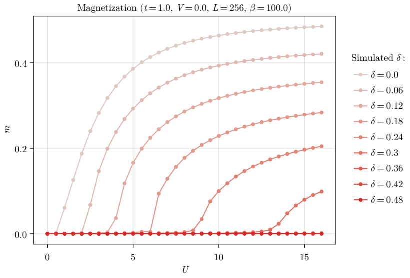


Figure 4: Numerical results for m.

Half-filling restriction

At any finite filling ($\delta \neq 0$) the commensurate AF phase is **unstable**. We limit to $\delta = 0$ (half-filling).

Mott localisation in the AF phase

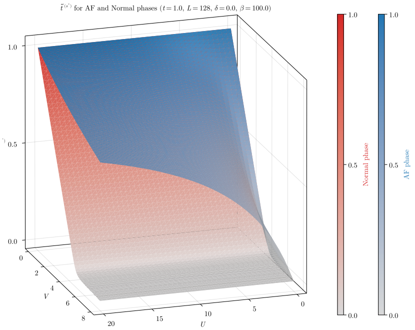


Figure 5: Comparative plot of the converged values for $\tilde{t}^{(s^*)}$ for the normal phase and the AF phase.

Setting $V > 0$, we get the new HFP $u^{(s^*)}$ as for the normal phase,

$$t \to \tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)} V}{2} .$$

Theoretical constraint:

$$0 < \tilde{t}^{(s^*)} < t$$

$$\Downarrow$$

Mott localisation in the AF phase

The local repulsion U **opposes bands flattening** $\implies$ counter-intuitive!

Magnetisation is **enhanced** by Mott localisation:

$$0 < \tilde{t}^{(s^*)} < t \quad \Longrightarrow \quad U/\tilde{t}^{(s^*)} > U/t\,.$$

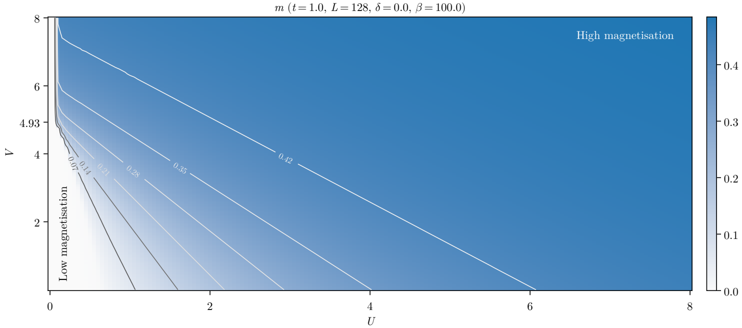


Figure 6: Numerical results for the HFP m. The gray curves indicate isomagnetic lines.

The HF solution in the commensurate AF phase

The **AF phase**, summarised:

- Breaks spin rotation and translational invariance;
- The HF approximation is only stable at half-filling;
- Presence of V-induced Mott localisation and **bands flattening**, enhanced by V and opposed by U;
- Two ways of magnetising the lattice: increasing U ("conventional") or using Mott effects ("unconventional");
- (Not shown in slides) always energetically favoured compared to the normal phase.

The superconducting phase

The translationally invariant SC phase

SC phase: breaking of charge conservation $\mathrm{U}(1)$, insurgence of a SC gap.

$$\Downarrow$$

Anomalous densities can be non-zero,

$$\Delta^{++}_{\alpha\beta} = \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \neq 0\,.$$

HFPs of the SC phase

Harmonics of the conventional density,

$$u_{\mathbf{k}} \equiv \left\langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \right\rangle \quad \to \quad u^{(\gamma)}\,.$$

Harmonics of the anomalous density,

$$w_{\mathbf{k}} \equiv \left\langle \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \right\rangle \quad \to \quad w^{(\gamma)}\,.$$

Gap symmetries in the singlet channel

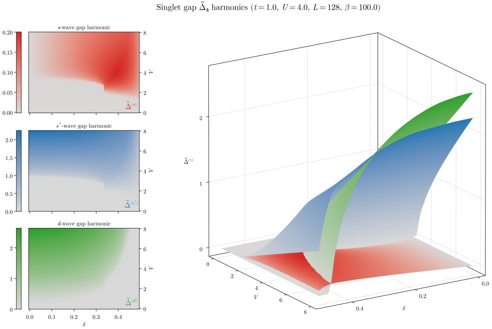


Figure 7: Singlet gap harmonics $\tilde{\Delta}^{(\gamma)}$, with $\gamma = s, s^*, d$ at $U/t = 4$.

Gap components in **singlet** channel:

$$\tilde{\Delta}^{(s)} = -Uw^{(s)}\,,$$

$$\tilde{\Delta}^{(s^*)} = Vw^{(s^*)}\,,$$

$$\tilde{\Delta}^{(d)} = Vw^{(d)}\,.$$

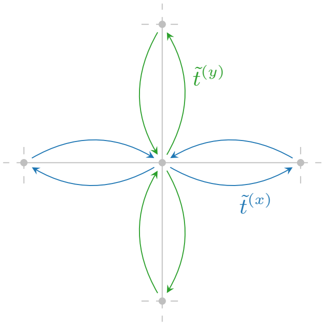
Internal constraint among singlet harmonics

$$V\sum_{\gamma\neq s}\left|w^{(\gamma)}\right|^2 \geq U\left|w^{(s)}\right|^2.$$

The nematic order parameter

In the SC phase, the renormalisation of t gives two components,

$$t \quad \rightarrow \quad \tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)}V}{2} \quad \text{and} \quad \tilde{t}^{(d)} \equiv -\frac{u^{(d)}V}{2} \, .$$



Anisotropic hopping

Direction-resolved effective hopping:

$$\tilde{t}^{(x)} \equiv \frac{\tilde{t}^{(s^*)} + \tilde{t}^{(d)}}{2} \, ,$$
$$\tilde{t}^{(y)} \equiv \frac{\tilde{t}^{(s^*)} - \tilde{t}^{(d)}}{2} \, .$$

$$\Downarrow$$

$\tilde{t}^{(d)} \neq 0 \implies$ **nematic SSB**.

Symmetry mixing and insurgence of nematicity

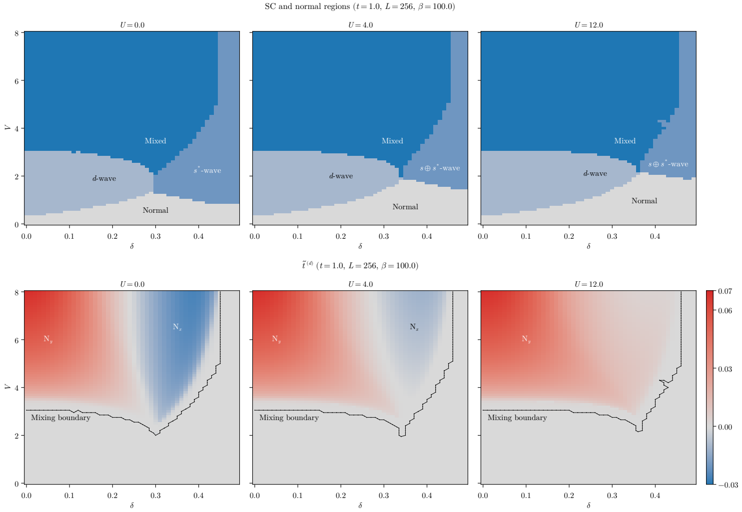


Figure 8: Comparison of phase diagrams for the singlet SC phase and convergence values for the nematic order parameter.

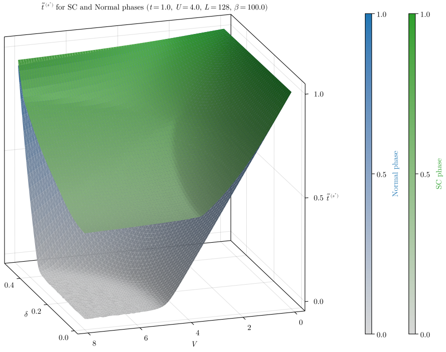


Figure 9: Comparative plot of the converged values for $\tilde{t}^{(s^*)}$ for the normal phase and the SC phase.

Isotropic hopping renormalisation:

$$\tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)}V}{2}\,.$$

Formally identical to normal and AF phases.

Isotropic flattening $\gg$ **Anisotropic correction**.

The HF solution in the SC phase

The **SC phase**, summarised:

- Breaks charge conservation and rotational symmetry;
- Symmetry mixing in singlet channel;
- Mott localisation and nematicity have the same origin;
- (Not shown in slides) always energetically favoured compared to the normal phase;
- (Not shown in slides) favoured at $V \gg U$ compared to the AF phase.

Conclusions and outlook

Our original results, summarised

The idea: we know that the EHM can host d-wave superconductivity. Can we obtain Mott physics?

At HF level, the non-local attraction:

- Acts as a source of Mott localisation;
- Enhances magnetisation at half filling;
- Can produce mixed-symmetry superconductivity;
- Can produce nematic superconductivity.

Further developments

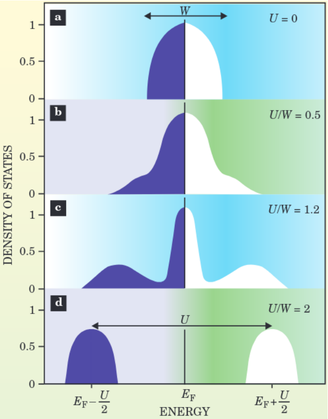
From here, we can follow various paths:

1. *Same model, other phases.*
   Triplet superconductivity, breaking of translational invariance, nematic antiferromagnetism...
2. *Expand the model.*
   Second nearest neighbours hopping, more refined potentials...
3. *More advanced numerical techniques.*
   DMRG for chains and ladders; hybrid DMFT (local sector) and HF (non-local sector) to retrieve Mott physics induced by U...

or any linear combination of the three!

That’s all, thank you!

Backup slides



Gabriel Kotliar and Dieter Vollhardt, 2004.
Physics Today 57 (March): 53–59.

As we increase the ratio U/t, the many-body DoS undergoes a transition:

(a) Free model, conventional DoS;
(b) Weak coupling, large coherent peak;
(c) Intermediate coupling, **narrow coherent peak**;
(d) Strong coupling, lower and upper Hubbard bands.

$$\Downarrow$$

Localisation = flattening of bare bands.

The iterative algorithm (iHF)

The HF solution is the **fixed point** of a non-linear map A,

$$\mathrm{A}(\mathbf{v}) = \mathbf{v}\,.$$

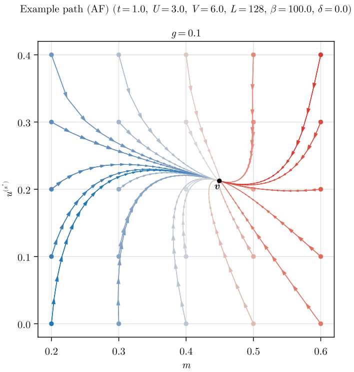
$$\Downarrow$$

Strategy: "follow" the map up to the fixed point,

$$\mathbf{v}_{i+1} \equiv g\mathrm{A}(\mathbf{v}_i) + (1-g)\mathbf{v}_i\,.$$

with $i > 0$, $g \in [0, 1]$.

$$\Downarrow$$



Example path for the AF phase.

Convergence criterion:

$$\forall j \quad : \quad [\mathbf{v}_{i+1} - \mathbf{v}_i]_j < \Delta v_j\,,$$

with $\Delta\mathbf{v}$ the vector of tolerances.